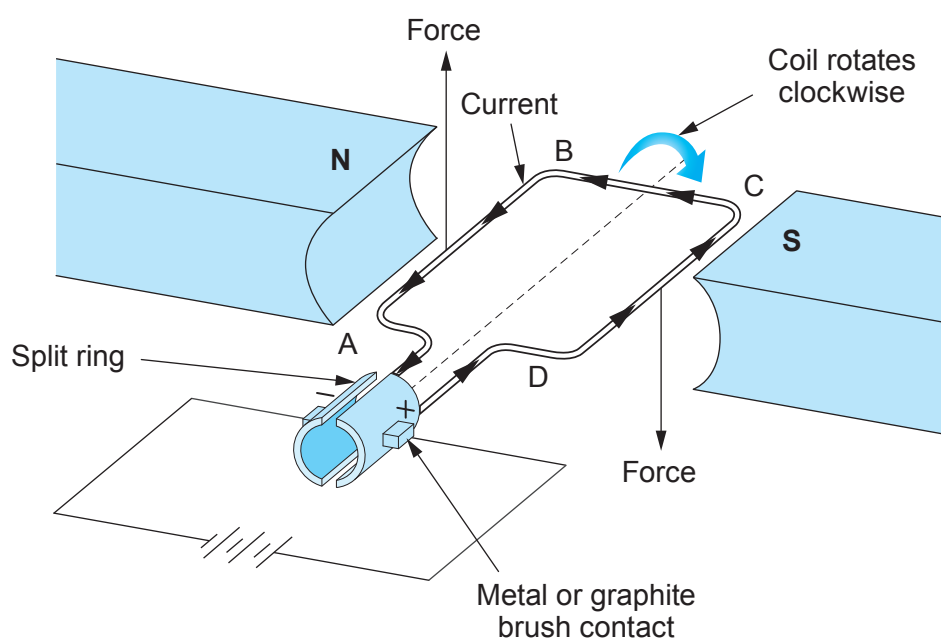


7. The diagram shows a simple d.c. electric motor consisting of a single turn of wire ABCD.



- (a) Describe how Fleming's left-hand rule can be used to show that the side **AB** of the wire moves upwards. [3]

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- (b) By considering the current in **AB**, explain how the split ring allows the coil to continue to rotate. [2]

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- (c) The sides AB and CD have lengths of 6.0 cm and the side BC has a length of 3.0 cm. The magnetic field has a strength of 0.04 T and a current of 0.7 A passes through the coil.

- (i) Use the equation:

$$\begin{array}{l} \text{force on a conductor (at right} \\ \text{angles to a magnetic field)} \\ \text{carrying a current} \end{array} = \begin{array}{l} \text{magnetic field} \\ \text{strength} \end{array} \times \text{current} \times \text{length}$$

to calculate the force on side **AB** of the coil.

[3]

Force = N

- (ii) Describe the effect on the force, on side **AB** of the coil, of doubling **both** the battery voltage **and** the number of coils.

[2]

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